Customer Churn Prediction

Capstone Project

Business report

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1. Understanding the Business Problem

Business Context

AlphaCom is a telecommunications provider experiencing customer churn, which can affect revenue and long-term customer relationships. Customer churn may be associated with factors such as service usage, billing preferences, contract types, and customer demographics. The existing retention approach is primarily reactive, making it difficult to identify customers who are at risk of leaving before they churn. Therefore, there is a need to analyse customer-level data to understand the factors associated with churn and identify customers who may be at higher risk.

Objective of the Study

The primary objective of this project is to develop a Customer Churn Prediction model that can identify customers who are likely to churn and understand the key factors associated with customer churn. The model aims to support a more proactive and data-driven customer-retention approach, enabling AlphaCom to prioritise customers who may require retention efforts and take timely action. This can provide an opportunity to improve customer retention, potentially reduce churn-related revenue losses, and improve customer lifetime value.

2. Dataset Overview

The dataset provided contains **12,055** customer records and **20 variables**, covering demographic information, tenure, subscribed services, contract and billing details, and customer churn status. Churn is the target variable indicating whether a customer has left the company

Variable	Description
Gender	Customer's gender
SeniorCitizen	Indicates whether the customer is a senior citizen
Partner	Indicates whether the customer has a partner
Dependents	Indicates whether the customer has dependents
Tenure	Number of months the customer has been with the company
PhoneService	Indicates whether the customer has phone service
MultipleLines	Indicates whether the customer has multiple phone lines
InternetService	Type of internet service subscribed to
OnlineSecurity	Indicates whether the customer has online security

Variable	Description
OnlineBackup	Indicates whether the customer has online backup
DeviceProtection	Indicates whether the customer has device protection
TechSupport	Indicates whether the customer has technical support
StreamingTV	Indicates whether the customer has streaming TV
StreamingMovies	Indicates whether the customer has streaming movie services
Contract	Type of customer contract
PaperlessBilling	Indicates whether paperless billing is enabled
PaymentMethod	Customer's payment method
MonthlyCharges	Monthly amount charged to the customer
TotalCharges	Total amount charged to the customer
Churn	Indicates whether the customer has churned

Table 1: Data Dictionary

Initial Data Quality Assessment

Data Type	No. of Variables
Object	18
Integer	1
Float	1

Table 2: Distribution of Variables by Data Type

- Some variables such as MonthlyCharges and TotalCharges required further data-type validation before they could be used for numerical analysis.

Statistical Summary

A statistical summary was performed for the numerical variables to understand their central tendency and spread. The dataset contains two numerical variables, SeniorCitizen and tenure. SeniorCitizen is a binary variable, while tenure represents the number of months a customer has been associated with AlphaCom.

The tenure variable has a mean of 31.24 months and a median of 28 months, indicating that the average customer has been with the company for approximately 31 months. The values range from -3 to 74 months, with the negative values requiring further investigation as potential anomalies during data preprocessing.

Duplicate Records

A duplicate-value check identified 2 duplicate records in the dataset. These records were removed to ensure that repeated observations did not influence the subsequent analysis.

Missing Values

The missing-value assessment identified 604 missing values in the tenure variable, while no missing values were observed in the remaining variables. The affected records contained information for the other customer attributes, with only the tenure value missing. The missing tenure values were therefore retained for further review and appropriate treatment during data preprocessing.

3. Data Preprocessing (Initial data cleaning)

Anomalous Value Check for Tenure

Statistic	Tenure (months)
Count	11,449
Mean	31.24
Standard Deviation	25.03
Minimum	-3
25th Percentile	6
Median	28
75th Percentile	54
Maximum	74

Table 3: Statistical summary of table

Observation:
The statistical summary indicates that the average customer tenure is 31.24 months, while the median tenure is 28 months. The observed values range from -3 to 74 months. The negative values are logically inconsistent with the definition of tenure and were therefore flagged for further investigation.

- The tenure variable contained 127 negative values, which are logically invalid because tenure represents the number of months a customer has been associated with AlphaCom. These values were replaced with the median tenure of the corresponding contract group, calculated using only valid, non-negative tenure values.
- Contract-wise median imputation was used because tenure may vary across different contract types. This approach preserves variation in tenure across customer contract groups rather than applying a single overall median.
- Following the treatment, the minimum tenure value became 0 months, and no negative tenure values remained.

4. Data Cleaning and Consistency Checks

- Several data-quality checks were performed to ensure that the customer information was consistent and suitable for analysis. The checks focused on the Churn, PaymentMethod, PhoneService–MultipleLines, and InternetService-related service variables.

Standardisation of Categorical Information

The Churn and PaymentMethod variables contained multiple representations of the same categories because of differences in capitalisation and additional spaces. These variations were standardised to ensure that the same category was treated consistently during analysis.

Variable	Issue Identified	Treatment	Result
Churn	8 different representations of Yes/No	Standardised capitalisation and removed extra spaces	2 consistent categories: Yes, No
PaymentMethod	Variations in capitalisation and extra spaces	Standardised capitalisation and removed extra spaces	4 consistent payment categories

Table 4 : Standardisation of Categorical Information

Observation:
The standardisation ensured that identical customer responses were grouped together correctly, preventing different spellings or formatting from being treated as separate categories.

Logical Consistency Checks

The analysis checked whether the information recorded for PhoneService and MultipleLines was logically consistent. A total of 220 potentially inconsistent records were identified.

Inconsistent Combination	Records
PhoneService = Yes & MultipleLines = No phone service	152
PhoneService = No & MultipleLines = Yes	68
Total	220

Table 5: Showing Phone Service and Multiple Lines Inconsistencies

These records were corrected based on the logical relationship between the two variables. After treatment, no inconsistent combinations remained.

Internet Service Consistency

A similar consistency check was performed between InternetService and the related services: OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, and StreamingMovies.A total of 357 potentially inconsistent records were identified across these six service variables.

Related Service	Potentially Inconsistent Records
TechSupport	66
DeviceProtection	64
OnlineBackup	63
OnlineSecurity	60
StreamingTV	54
StreamingMovies	50
Total	357

Table 6 : Internet Service Consistency

5. Data Cleaning Outcome

The consistency checks and subsequent corrections resulted in a more standardised dataset for further analysis.

Data Quality Area	Before Treatment	After Treatment
Churn categories	8 representations of Yes/No	2 consistent categories
PaymentMethod	Multiple formatting variations	4 consistent categories
PhoneService–MultipleLines	220 inconsistent records	0 remaining
InternetService-related services	357 inconsistent records	0 remaining

Table 7 : Data cleaning outcome

Business implication:
Standardising customer information and resolving logical inconsistencies improves the reliability of subsequent customer analysis and helps ensure that churn patterns are based on consistent customer records rather than differences in data entry or formatting.

Data Type and Currency Standardisation

Methodology

The MonthlyCharges and TotalCharges variables contained monetary values in different currency formats, along with missing-like values such as $nan and £nan. To ensure consistency, the monetary values were converted to USD using the applicable exchange rates, after which the currency symbols were removed and both variables were converted from text to numerical format (float64). Missing values were retained for subsequent treatment.

Findings

The initial check identified 301 non-numeric values in MonthlyCharges and 1,205 non-numeric values in TotalCharges. These included currency symbols and missing-like entries.

Variable	Non-Numeric Values
MonthlyCharges	301
TotalCharges	1,205

Table 8: Non-Numeric Values in Monetary Variables

After currency standardisation and conversion, both variables were successfully converted to numerical format.

Business implication: Standardising monetary values ensures that customer charges can be compared consistently and used reliably in subsequent analysis and modelling.

Statistic	**MonthlyCharges ($)**	**TotalCharges ($)**
Count	11,752	10,848
Mean	64.37	2,292.07

Standard Deviation	30.33	2,277.54
Minimum	15.29	-197
Median	71.36	1,329.18
Maximum	121.67	9,039.92

Table 9: Statistical Summary of Monetary Variables

Observation:

Monthly charges range from $15.29 to $121.67, with an average of $64.37. Total charges show greater variation, with values ranging from -$197.00 to $9,039.92. The negative minimum value in TotalCharges prompted the anomaly investigation and treatment described above**.**

Treatment of Negative Total Charges

Methodology

After converting TotalCharges into numerical format, the variable was examined for anomalous values. Since TotalCharges represents the cumulative charges associated with a customer, negative values were considered logically invalid. The identified records were further reviewed based on contract type, churn status, and tenure to understand whether a valid business explanation could be established.

Findings

A total of 147 negative TotalCharges records were identified, ranging from -$197.00 to -$0.10.

The negative values occurred across all three contract types and were present among both churned and non-churned customers.

The available variables did not provide sufficient evidence of a valid business reason, such as refunds or credits, for these negative cumulative charges. Therefore, the values were treated as anomalous.

The 147 negative TotalCharges values were replaced with the median of valid, non-negative TotalCharges observations ($1,373.05).

Following treatment:

- Negative TotalCharges values remaining: 0
- TotalCharges remained in numerical format.
- No negative cumulative charges remained in the dataset.

Business implication: Treating the anomalous negative charges prevents unrealistic billing values from influencing the analysis and subsequent churn prediction model.

6. Missing Value Treatment

Methodology

Missing values were identified in three numerical variables: tenure, MonthlyCharges, and TotalCharges. Since these variables represent numerical customer and billing information, median imputation was used to retain the affected customer records while reducing the influence of unusually high or low values.

Variable	Missing Values Before Treatment	Missing Values After Treatment
Tenure	604	0
MonthlyCharges	301	0
TotalCharges	1,205	0

Table 10: Missing Values Before and After Treatment

The tenure variable had 604 missing observations, which were replaced with the median tenure of 28 months. Similarly, 301 missing MonthlyCharges values and 1,205 missing TotalCharges values were replaced using their respective median values.

After treatment, all three variables had zero missing observations, confirming that the missing-value treatment was completed successfully.

Business implication: Median imputation allowed the affected customer records to be retained for analysis and modelling while avoiding the introduction of extreme values that could distort customer-level patterns.

7. Final Data Quality Check

After completing the data-cleaning and preprocessing steps, a final quality check was performed to ensure that the dataset remained consistent and suitable for further analysis. The check reviewed the cleaned dataset for any records that appeared identical after standardisation and confirmed whether they represented genuine duplicate observations.

Some records appeared identical after the cleaning process; however, these were reviewed and were not treated as additional duplicate customer records. Therefore, no further records were removed at this stage.

The final validation also confirmed that the key data-quality issues identified during preprocessing had been appropriately addressed, including:

- No missing values remained in the treated numerical variables.
- No negative tenure values remained.
- No negative TotalCharges values remained.
- No logical inconsistencies remained between the related service variables.
- The categorical values had been standardised.

- The dataset was ready for the subsequent modelling preparation steps.

8. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to understand the distribution of customer churn and identify customer, service, contract, and billing characteristics associated with churn. The analysis focused on identifying meaningful patterns that could help AlphaCom understand its customer base and support the subsequent churn prediction modelling.

Univariate Analysis

Univariate analysis was performed to understand the distribution and characteristics of individual variables in the dataset.

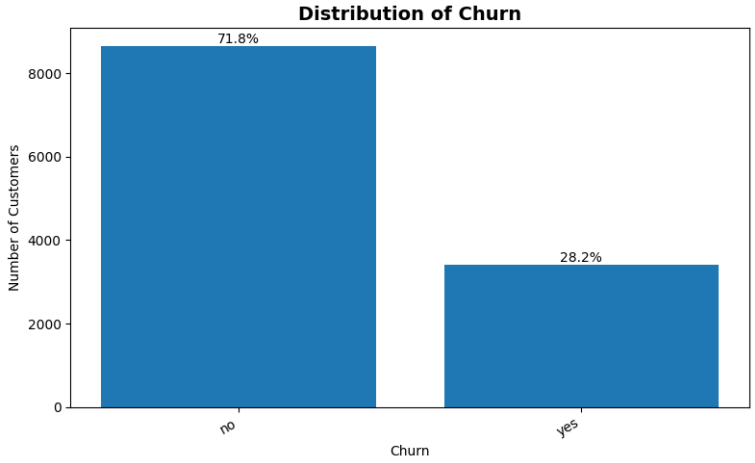


Figure 1: Distribution of Customer Churn

- The dataset shows that 71.8% of customers have not churned, while 28.2% have churned. This indicates that although the majority of customers remain with AlphaCom, a substantial proportion of the customer base has left the company.
- The 28.2% churn rate highlights the need to identify customers who may be at risk of leaving and supports the need for a proactive customer-retention approach.

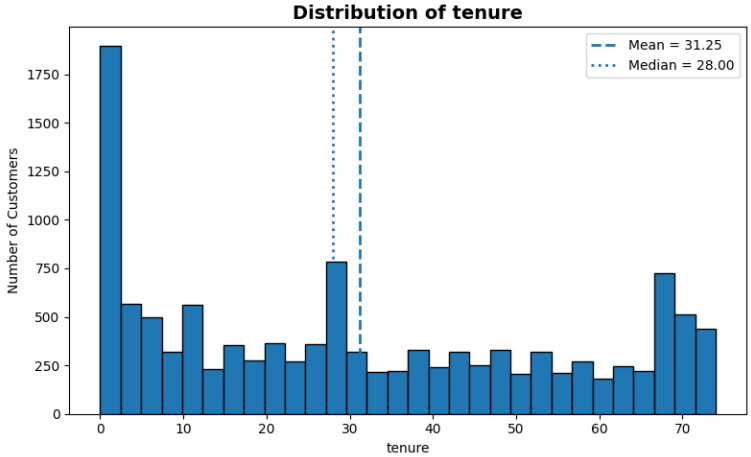


Figure 2: Distribution of Customer Tenure

- The plot shows that AlphaCom has customers at different stages of their rel ationship with the company, from newly acquired customers to those who have been with the company for several years. Customer tenure ranges from 0 to 74 months, with an average tenure of approximately 31 months.
- A considerable number of customers have relatively short tenures, while a substantial customer base has longer relationships with AlphaCom. This highlights the importance of understanding customer behaviour at different stages of the customer relationship, particularly to identify when customers may be more likely to leave

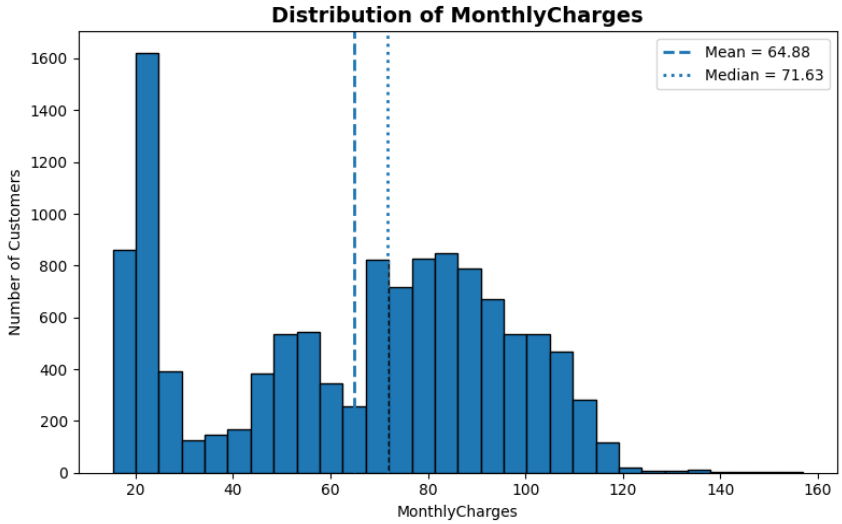


Figure 3: Distribution of Monthly Charges

- The plot shows that AlphaCom customers have monthly charges ranging from approximately $15 to $160, with an average monthly charge of around $64.88.A
- A large number of customers have relatively lower monthly charges, while many customers also fall within the $70–$100 range. This indicates that customers have different levels of monthly spending based on their subscribed services and plans.

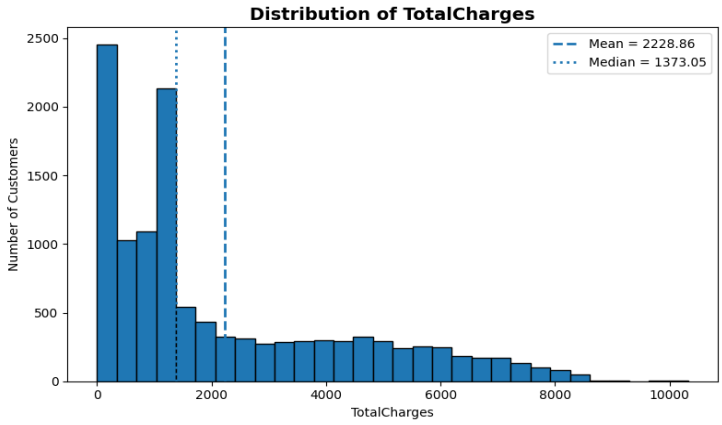


Figure 4: Distribution of Total Charges

- The plot shows that the total amount charged to customers ranges from approximately $0 to $10,000, with an average of around $2,228.86. A large number of customers have relatively lower total charges, while customers with longer relationships with AlphaCom tend to accumulate higher charges over time. This reflects differences in both customer tenure and service spending.
- Understanding total charges can therefore help AlphaCom assess the value of its customer relationships and examine whether customers contributing different levels of revenue show different churn patterns.

Bivariate analysis

Bivariate analysis was performed to understand how customer characteristics, service preferences, contract terms and billing-related factors are associated with customer churn. The analysis focuses on differences in churn behaviour across customer groups to identify areas that may require greater attention from AlphaCom

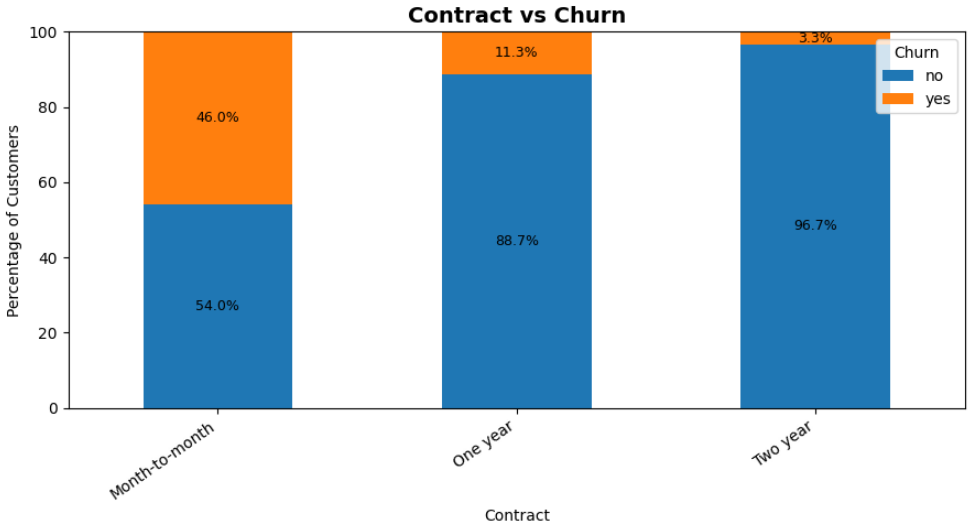


Figure 5: Contract Type and Customer Churn

- The analysis shows a clear difference in churn across contract types. 46.0% of month-to-month customers have churned, compared with 11.3% of one-year contract customers and only 3.3% of two-year contract customers.
- In contrast, 96.7% of two-year customers and 88.7% of one-year customers remained with AlphaCom. This indicates that customers with shorter contract commitments show substantially higher churn, while customers with longer-term contracts tend to remain with the company.
- Contract type is strongly associated with customer retention, with month-to-month customers representing the group with the highest observed churn. This group may therefore require greater attention in AlphaCom's retention efforts

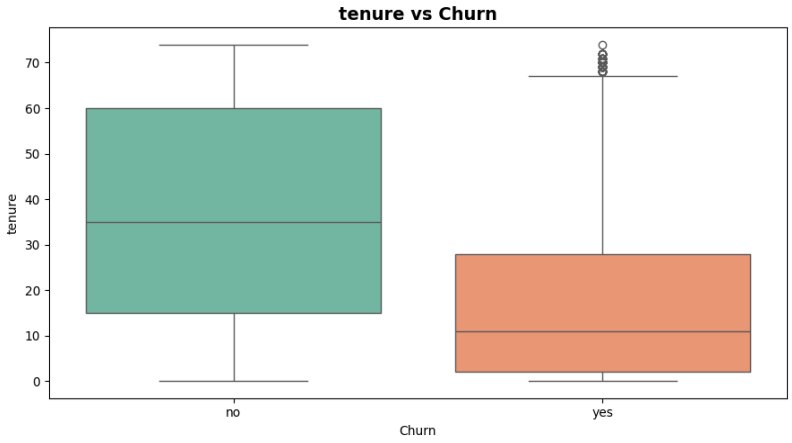


Figure 6 : Customer Tenure and Churn

- The analysis shows a clear difference in tenure between customers who stayed with AlphaCom and those who churned. Customers who remained with the company have a median tenure of approximately 35 months, while customers who churned have a much lower median tenure of approximately 11 months.
- This indicates that churn is more common among customers with shorter relationships with AlphaCom, whereas customers who have stayed longer tend to have more established relationships with the company.
- Customers with shorter tenure show a higher observed level of churn, highlighting the importance of engaging and retaining customers during the early stages of their relationship with AlphaCom.

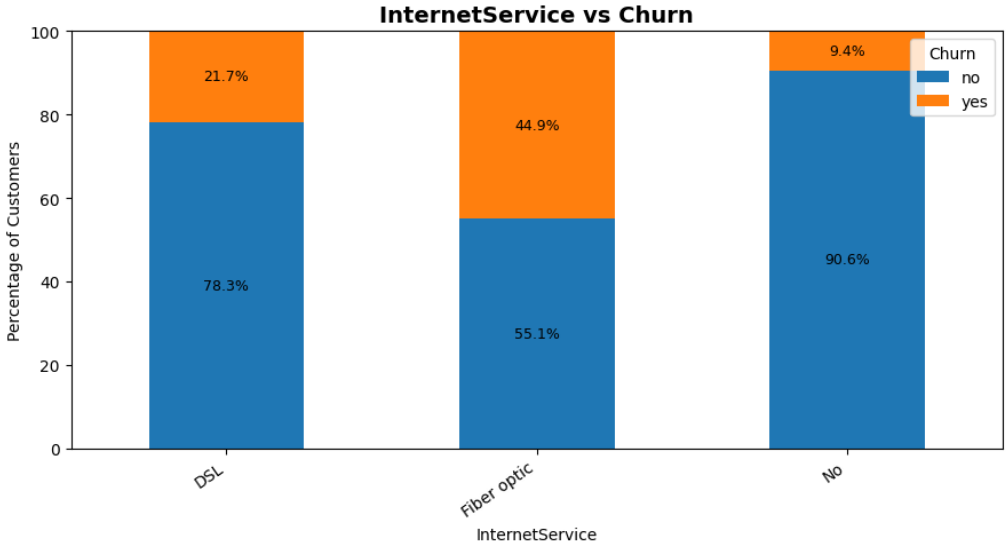


Figure 7: Internet Service and Customer Churn

- The analysis shows noticeable differences in churn across internet service types. 44.9% of customers using Fiber optic services have churned, compared with 21.7% of DSL customers and 9.4% of customers without internet service.
- In contrast, 90.6% of customers without internet service remained with AlphaCom. This indicates that customer churn varies considerably across internet service groups

and highlights the need to understand the experience of customers using different types of internet services.

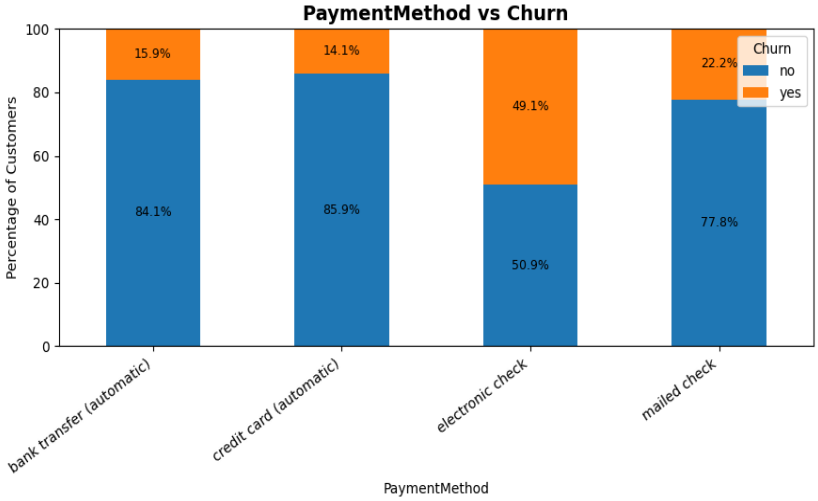


Figure 8: Payment Method and Customer Churn

- The analysis shows clear differences in churn across payment methods. 49.1% of customers using electronic checks have churned, compared with 22.2% of customers using mailed checks, 15.9% using bank transfers, and 14.1% using credit cards.
- In comparison, the majority of customers using other payment methods remained with AlphaCom. This indicates that customer churn varies across payment groups and that payment behaviour may be associated with customer retention.
- Customers using electronic checks show the highest observed churn, making this group important to examine further when understanding the factors associated with customer churn.

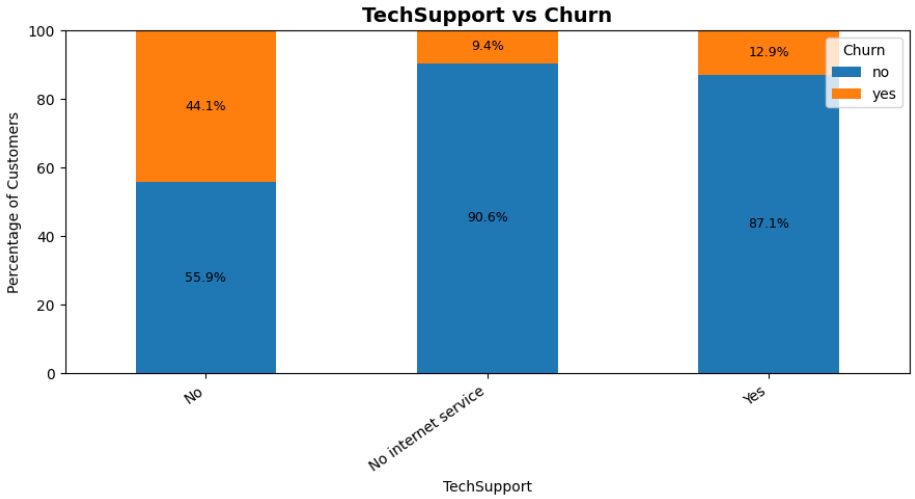


Figure 9: Technical Support and Customer Churn

- The analysis shows a noticeable difference in churn between customers based on technical support availability. 44.1% of customers without technical support have churned, compared with 12.9% of customers who have technical support.

- Among customers with no internet service, the churn rate is 9.4%. This indicates that customers without technical support show a considerably higher level of churn than customers who have access to this service suggesting that access to support services may be an important factor to consider when understanding customer retention.

Multivariate Analysis

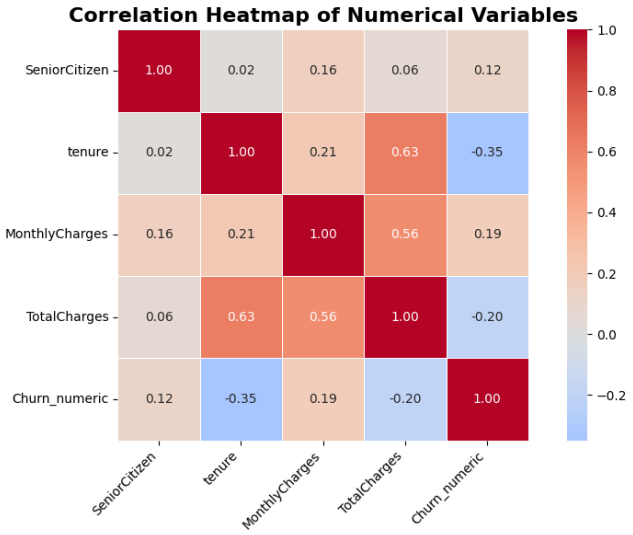


Figure 10: Correlation Between Numerical Variables and Churn

- The correlation analysis shows that tenure has the strongest relationship with churn among the numerical variables, with a correlation of -0.35, indicating that customers with longer tenure tend to show lower churn.
- Total charges have a weaker negative relationship with churn (-0.20), while monthly charges show a weak positive relationship (0.19). Senior citizen status has a relatively weak relationship with churn (0.12).

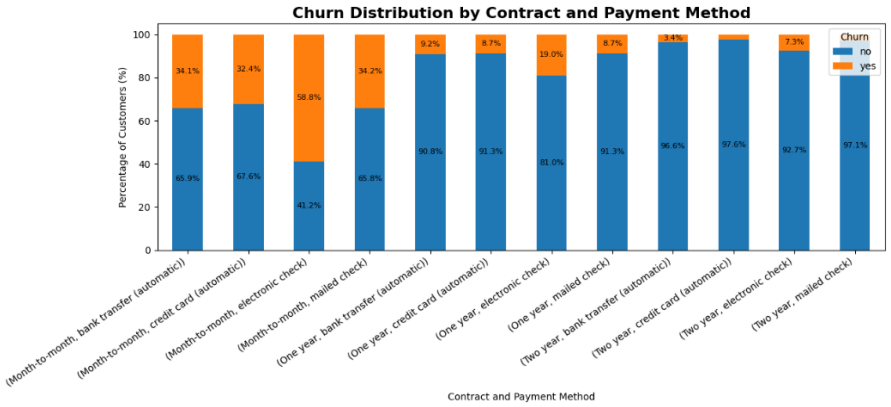


Figure 11: Customer Churn by Contract Type and Payment Method

- The analysis shows that month-to-month customers have higher churn across payment methods, with the highest observed churn of 58.8% among customers using electronic checks. For one-year contracts, churn is considerably lower, ranging from

8.7% to 9.2%, while two-year contracts show the lowest churn, ranging from 3.4% to 8.7% across payment methods.

- Customers on shorter contracts, particularly month-to-month customers using electronic checks, represent an important group to consider when understanding and addressing customer churn.

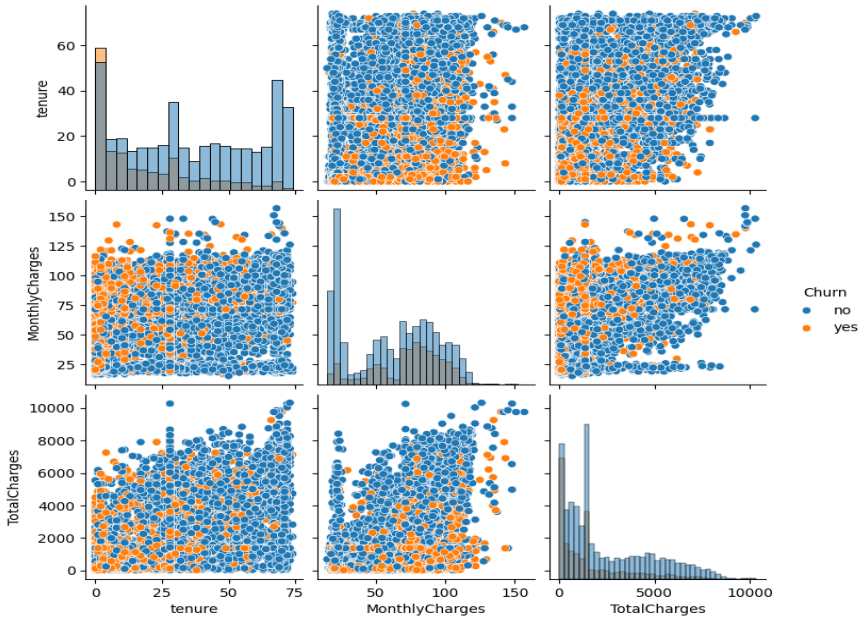


Figure 12 : Pairplot

- The pairplot shows that TotalCharges generally increase as tenure increases, which is expected because customers who stay longer accumulate more charges.
- MonthlyCharges also show a positive relationship with TotalCharges, while the churned customers (orange) appear more frequently among customers with shorter tenure and lower total charges. However, there is considerable overlap between churned and retained customers, indicating that no single numerical variable clearly separates the two groups.

Insights/observations based on EDA

Customers on month-to-month contracts show a higher tendency to churn compared with customers on longer-term contracts, indicating that contract duration is an important factor associated with churn.

- Customers using electronic check show a relatively higher churn tendency, making payment method an important variable for understanding customer attrition.
- Customers with shorter tenure are more likely to churn, suggesting that the early stage of the customer relationship is particularly important for retention.
- Churn varies across internet service categories, indicating that the type of service subscribed to may influence customer retention.

- Customers with higher monthly charges show a greater tendency towards churn, suggesting that pricing or perceived value may be relevant to customer retention.
- Variables such as Online Security, Tech Support and Online Backup show differences in churn behaviour, indicating that service features and support may be associated with customer retention.
- **Overall observation:** The EDA indicates that churn is influenced by a combination of contract type, tenure, payment method, service-related factors and customer charges, rather than a single variable. These observations were considered while preparing the data and developing the churn prediction models

Note: A comprehensive Exploratory Data Analysis (EDA), including univariate, bivariate, and multivariate analysis, was performed and documented in the Google Colab notebook The Business Report presents only the key analyses and findings that are directly relevant to the business problem of understanding and predicting customer churn.

10. Data Preparation – Outlier Treatment

Outlier Assessment

- After completing the EDA, the numerical variables were reviewed for potential outliers using the IQR method. The assessment showed no potential outliers in tenure or MonthlyCharges, while 75 potential outliers were identified in TotalCharges.
- The 75 potential outliers in TotalCharges were reviewed rather than removed automatically. Since TotalCharges represents cumulative customer charges, higher values can naturally occur for customers with longer tenure or higher monthly charges. Therefore, the observations were retained as they may represent genuine high-value customers rather than data errors.
- Key takeaway: The outlier assessment ensured that unusual charge values were reviewed before modelling while avoiding the unnecessary removal of potentially valid customer records.

10. Model Building and Evaluation

Model Preparation

The target variable, Churn, contains two customer groups: customers who did not churn and customers who churned. Approximately 71.8% of customers did not churn, while 28.2% churned, indicating an imbalance between the two groups. Since identifying customers at risk of churn is an important part of the business objective, this imbalance was considered during model preparation.

Class Balancing Using SMOTE

To provide the models with a more balanced representation of both customer groups, SMOTE (Synthetic Minority Over-sampling Technique) was applied to the training data only. After applying SMOTE, the training data contained an equal representation of churned and non-churned customers. The test data was not modified so that model performance could be evaluated on the original customer distribution.

Train–Test Data Preparation and Data Leakage Prevention

The dataset was divided into training and test sets before applying model-specific transformations. Feature scaling was fitted using only the training data and then applied to the test data. Similarly, SMOTE was applied only to the training data. The test set was kept unchanged and wa s used only for final model evaluation. This approach ensured that information from the test data did not influence model training or evaluation.

Model-Ready Dataset

Following data cleaning, anomaly treatment, train–test splitting, feature scaling, and class balancing, the training data was prepared for model development. The test data remained untouched by SMOTE and was reserved for evaluating how well the models perform on unseen customer records.

Baseline Logistic Regression

Logistic Regression was developed as the baseline classification model to predict whether a customer is likely to churn or remain with AlphaCom. The cleaned dataset was divided into training and testing sets before applying modelling-specific transformations. Categorical variables were converted into numerical format, resulting in 23 predictor variables. Since the training data had an imbalance between churned and non-churned customers, SMOTE was applied only to the training data to provide a balanced representation of both classes. The Logistic Regression model was then trained on the balanced training data, while the test data was kept unchanged for unbiased evaluation.

Model Fit

The Logistic Regression model successfully converged and produced a Pseudo R-squared of 0.333. The overall model was statistically significant, with an LLR p-value below 0.001, indicating that the set of predictors provides useful information for distinguishing between customers who churn and those who do not. The model was therefore used as the baseline against which subsequent machine learning models can be compared.

Model Performance Evaluation

Since the primary business objective is to identify customers who are likely to churn so that AlphaCom can take timely retention actions, Recall was treated as an important evaluation metric. A False Negative occurs when an actual churner is incorrectly predicted as a non-churner, which could result in a missed opportunity for customer retention. False Positives are also relevant because they may lead the company to target customers who were not actually

going to churn. Therefore, Recall, Precision, F1-score, Accuracy and ROC-AUC were considered together to provide a balanced view of model performance.

The baseline Logistic Regression model achieved an Accuracy of 75.28%, Recall of 79.84%, Precision of 52.81%, F1-score of 63.57%, and ROC-AUC of 84.87% on the test dataset. The model correctly identified 780 actual churn customers, while 197 churners were incorrectly classified as non-churners. The Recall of 79.84% indicates that the model identified approximately 80% of customers who actually churned. However, the Precision of 52.81% shows that a considerable number of customers predicted as churners did not actually churn. Overall, these results provide the baseline performance for evaluating the subsequent models developed in the project.

Accuracy	Recall	Precision	F1-Score
75.28%	79.84%	52.81%	63.57%

Table 11: Model Performance

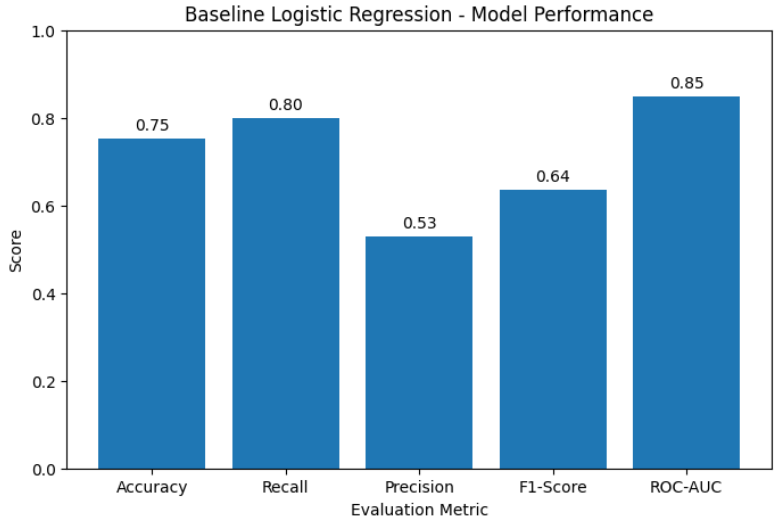


Figure 13: Model Performance

Confusion Matrix Interpretation

The confusion matrix was examined to understand the types of prediction errors made by the model. The model correctly classified 1,942 non-churn customers and 780 churn customers. It incorrectly classified 697 non-churn customers as churners (False Positives) and 197 actual churners as non-churners (False Negatives). This analysis helps assess not only how accurate the model is overall, but also how effectively it identifies customers who are at risk of leaving

11. Detecting and Dealing with Multicollinearity

Multicollinearity was checked using the Variance Inflation Factor (VIF) before building the final Logistic Regression model. Most variables had VIF values below 5. MonthlyCharges had the highest VIF of 7.27, followed by InternetService_No at 4.25 and TotalCharges at 2.63. Since no variable had a VIF of 10 or above, no variable was removed solely based on VIF.

The Logistic Regression model was then fitted and the statistical significance of the predictors was evaluated using p-values. Variables with high p-values were removed to simplify the model and retain statistically significant predictors. This reduced the number of predictors from 23 to 18 variables for the final Logistic Regression model.

12. Final Logistic Regression Model

The Logistic Regression model was refitted using the 18 selected variables. SMOTE was applied to the training data to address the class imbalance, while the test data was kept unchanged. StandardScaler was fitted on the training data and the same transformation was applied to the test data. The final training dataset contained 12,020 observations and 18 features, while the test dataset contained 3,616 observations and the same 18 features.

The final Logistic Regression model achieved a test accuracy of 71.0%, recall of 87.2%, precision of 47.3%, and F1-score of 61.3%. The ROC-AUC was 0.849, indicating that the model has good ability to distinguish between customers who churn and those who do not.

ROC Curve

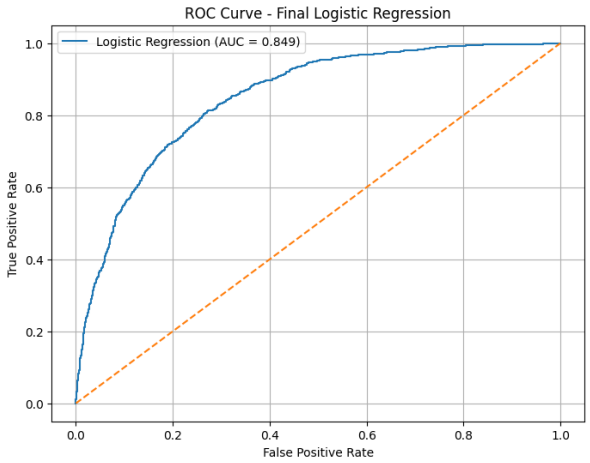


Figure 14:ROC Curve

The ROC curve of the final Logistic Regression model produced a ROC-AUC of 0.849. The curve remained substantially above the random-classification reference line, indicating good discriminatory ability. This shows that the model can effectively rank customers according to their likelihood of churn.

Precision–Recall Curve

The Precision–Recall curve was examined because identifying churners is particularly important for AlphaCom. The model achieved an Average Precision of 0.668. The curve demonstrates the expected trade-off between precision and recall: as the model attempts to identify more potential churners, the number of false-positive predictions also increases.

Selection of Classification Threshold

The default classification threshold of 0.50 was further evaluated to identify a more suitable balance between precision and recall. A threshold of 0.6097 was identified as the selected operating threshold.

At this threshold, the model achieved 58.5% precision, 71.8% recall, and an F1-score of 64.4%. The threshold was therefore used for the Logistic Regression model to obtain a more balanced identification of potential churn customers.

13. Model Building – Advanced Models

Methodology

Six advanced classification models were developed to improve churn prediction beyond the baseline Logistic Regression model: Decision Tree, Bagging, Random Forest, AdaBoost, Gradient Boosting and XGBoost. The models were trained on the prepared training dataset and evaluated on the unseen test dataset using Accuracy, Recall, Precision, F1-score and ROC-AUC. Since the primary business objective is to identify customers who are likely to churn, Recall for the churn class was given particular importance.

Model	Accuracy	Recall	Precision	F1-Score	ROC-AUC
Decision Tree	71.18%	55.68%	47.18%	51.08%	66.35%
Bagging	77.32%	61.00%	57.58%	59.24%	80.58%
Random Forest	78.10%	63.87%	58.70%	61.18%	82.91%
AdaBoost	75.25%	79.22%	52.80%	63.36%	84.78%
Gradient Boosting	77.82%	72.16%	57.09%	63.74%	85.15%
XGBoost	78.10%	62.33%	58.70%	60.60%	82.87%

Table 12: Test-Set Performance Comparison of Advanced Classification Models

Key observations:

- The Decision Tree identified 55.68% of actual churn customers, giving the lowest recall among the models.
- Bagging and Random Forest improved churn identification, with recall of 61.00% and 63.87%, respectively.
- AdaBoost achieved the highest recall at 79.22%, identifying the largest proportion of actual churn customers.
- Gradient Boosting achieved the highest ROC-AUC of 85.15% and an F1-score of 63.74%, while maintaining a recall of 72.16%.
- XGBoost and Random Forest achieved 78.10% accuracy, but their recall was lower than AdaBoost and Gradient Boosting.

- Based on these results, the comparatively stronger models were taken forward for hyperparameter tuning to check whether their performance could be further improved.

14. Model Performance Improvement Using Hyperparameter Tuning

Methodology:

Based on the initial test results, Random Forest, AdaBoost and Gradient Boosting were selected for hyperparameter tuning. These models showed comparatively stronger performance for identifying churn customers and had potential for further improvement. Hyperparameter tuning was performed to adjust model settings and improve their performance on unseen customers. The tuned models were again evaluated using Accuracy, Recall, Precision, F1-Score and ROC-AUC, with **Recall as the primary metric** because identifying customers likely to churn is the main objective.

Model	Accuracy	Recall	Precision	F1-Score	ROC-AUC
Random Forest	78.07%	65.92%	58.33%	61.89%	82.98%
AdaBoost	50.06%	97.75%	34.87%	51.40%	78.20%
Gradient Boosting	71.18%	85.36%	48.12%	61.55%	83.36%

Table 13: Test-Set Performance Comparison of Tuned Models

Key observations:

- **Random Forest:** Tuning produced only a small improvement in Recall, F1-Score and ROC-AUC. The model continued to show a large difference between training and test performance, indicating that overfitting remained.
- **AdaBoost:** Recall increased substantially from 79.22% to 97.75%, allowing the model to identify most actual churn customers. However, Precision fell to 34.87%, meaning many customers predicted as churners were actually non-churners. Its overall performance therefore became less balanced.
- **Gradient Boosting:** Recall improved from 72.16% to 85.36%, meaning more actual churn customers were identified. However, Precision and overall accuracy decreased, with 899 non-churn customers incorrectly classified as churn.

15. Model Performance Comparison and Final Model Selection

Methodology

The three tuned models—Random Forest, AdaBoost and Gradient Boosting—were evaluated on the test dataset using Accuracy, Recall, Precision, F1-Score and ROC-AUC. Since the primary

objective of AlphaCom is to identify customers who are likely to churn, Recall was considered the primary metric, while the remaining metrics were used to assess the overall balance of predictions.

- Tuned AdaBoost achieved the highest Recall at 97.75%, identifying the largest proportion of actual churn customers.
- Tuned Gradient Boosting achieved 85.36% Recall and the highest ROC-AUC among the tuned models at 83.36%.
- Tuned Random Forest achieved the highest Accuracy (78.07%) and Precision (58.33%) among the three tuned models, but its Recall was lower at 65.92%.
- As identifying potential churners is the main objective, Tuned AdaBoost was selected as the final model based on its highest Recall.

Since Recall was identified as the primary metric for the business objective of identifying customers at risk of churn, greater importance was given to the model's ability to minimise false negatives. Based on this criterion, Tuned AdaBoost was selected as the final model because it achieved the highest test Recall of 97.75%. The lower Precision and Accuracy were considered as trade-offs associated with achieving higher churn detection.

16. Final Model – Tuned AdaBoost

The final Tuned AdaBoost model achieved 97.75% Recall for the churn class. Its confusion matrix shows:

- True Positives (TP): 955 – actual churn customers correctly identified as churners.
- False Negatives (FN): 22 – actual churn customers incorrectly classified as non-churners.
- True Negatives (TN): 855 – actual non-churn customers correctly identified as non-churners.
- False Positives (FP): 1,784 – non-churn customers incorrectly classified as potential churners.

This means the model misses only 22 of the 977 actual churn customers, which explains its high Recall. However, the model also generates 1,784 false-positive predictions, meaning that a substantial number of non-churning customers were classified as potential churners, resulting in a lower Precision of 34.87% and Accuracy of 50.06%.

From a business perspective, the model is therefore designed to capture as many potential churners as possible, but AlphaCom should be aware that some customers identified for retention may not actually churn. This trade-off should be considered when planning retention campaigns and allocating resources.

17. Key Insights

- Contract type is an important factor associated with customer churn. Customers across different contract categories may have different retention needs and should be considered when designing retention strategies.
- Payment method is associated with churn, with electronic check users identified as an important customer segment to monitor.
- Customer tenure is an important factor, indicating that customers at different stages of their relationship with AlphaCom may require different retention approaches.
- The churn prediction model provides AlphaCom with an opportunity to identify customers at risk of churn before they leave, enabling proactive retention actions.

Recommendations

- Provide personalised retention offers and suitable incentives to customers identified as being at high risk of churn.
- Review the electronic check payment experience and simplify billing and payment processes where possible.
- Strengthen onboarding and early-stage customer support to improve engagement among newer customers.
- Proactively contact customers identified as high-risk, understand their concerns, and provide appropriate solutions before cancellation.
- Use targeted retention strategies based on customer characteristics and predicted churn risk rather than applying the same offer to all customers.
- Regularly collect customer feedback and address billing, service, and support-related issues that may contribute to customer dissatisfaction.
- Use the churn prediction model to support proactive retention activities, with the objective of reducing avoidable customer losses, protecting revenue, and improving customer lifetime value.